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Wheel-Leg Collaborative Control for Wheel-legged Robots Based on MPC with Preview

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***Abstract*—In the field of wheel-legged robots (WLRs), two types of configurations exist: toe-wheeled and knee-wheeled. In the knee-wheeled configuration, the robot is capable of switching between leg walking and wheel driving modes. However, the participation of the calf leg in driving control is often limited during the switch to the wheel driving mode, which makes it similar to a rocker-arm suspension vehicle with an additional unsprung mass. The presence of redundant calf legs in knee-wheeled WLRs can increase the unsprung mass, potentially impacting the overall performance of the robot. To address this issue, this paper presents a collaborative control method based on road preview for knee-wheeled WLRs. The proposed approach simplifies the wheeled system to a quarter car model and designs a Model Predictive Controller (MPC). By effectively integrating the redundant calf legs into the control strategy, the proposed approach aims to mitigate the negative effects of increased unsprung mass and enhance the overall performance of the robot in terms of vertical comfort and road holding ability. Simulation results demonstrate that the vertical performance of the wheel leg system can be effectively improved through the control of the calf leg when traversing undulating terrains. The proposed anticipatory control method and collaborative controller were found to effectively enhance the overall performance of the robot in terms of comfort and road holding ability. The findings of this study provide insights into the development of control strategies for knee-wheeled legged robots and contribute to the advancement of wheeled legged locomotion research.**

***Index Terms*—Wheel-legged robots, Rocker-arm suspension, Knee-wheel, Active suspension, Wheel-leg collaboration**

I. Introduction

Recently, some advanced legged mobile robots (LMRs) have been designed to imitate their counterparts in the nature, like hexapod [1], BigDog [2], Cheetah [3], Octopus [4] et al. While the wheeled mobile robots (WMRs) have been widespread concerned for a long time for their relatively low mechanical complexity and ability to move quickly on the flat terrain.

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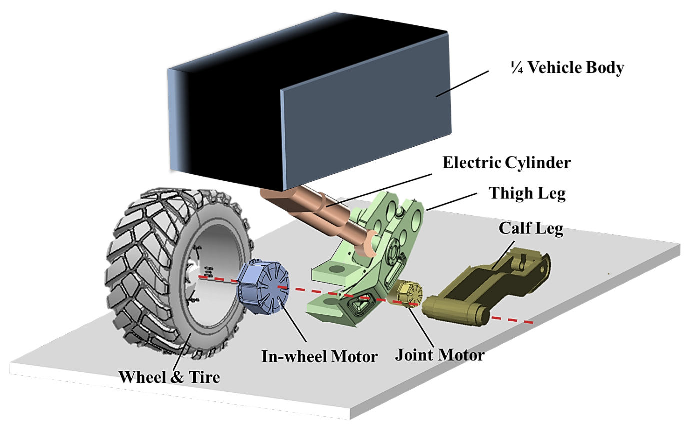

Fig. 1. Mechanical structure of our wheel-legged robot

Bridging the gap between legged and wheeled robots, wheel-legged robots (WLRs) have emerged as a promising solution [5], [6]. WLRs offer several advantages, including energy efficiency and environmental adaptability, by combining the high mobility of wheeled robots on roads with the high trafficability of legged robots on unstructured terrains [7], [8]. WLRs can be classified into two main types of mechanical constructions: toe-wheel, where wheels are placed at the tips of robot legs, and knee-wheel, where wheels are located at the knee joints of robot legs [9], [10].

Compared with toe-wheeled mechanism structure, knee-wheel construction offers the advantage of being able to switch between wheeled motion and legged motion modes based on different working conditions. In legged motion, the knee wheel has a smaller mechanical inertia, which can enhance agility and dynamic performance. However, in wheeled motion, both knee-wheeled and toe-wheeled WLRs have some limitations when evaluated based on the standards of WMRs or commercial vehicles. One such limitation is the relatively greater

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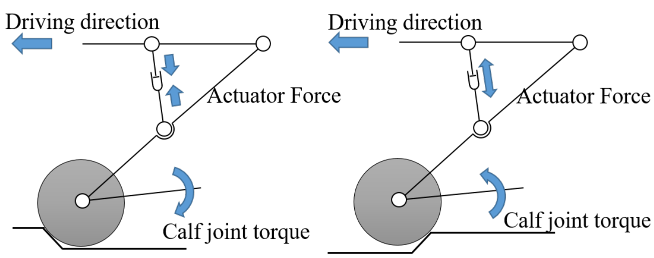

Fig. 2. Working principle of wheel-legged robot suspension control

unsprung mass associated with knee-wheel construction. It is well-known that a smaller ratio of unsprung mass to sprung mass results in better vehicle comfort, handling stability, acceleration, and braking performance on uneven roads [11]. Additionally, locking the calf leg in knee-wheeled WLRs can be energy-consuming, and not locking it may lead to interference and collision issues.

In our knee-wheeled WLR mechanical structure, as depicted in Fig. 1, a quarter-model of the WLR comprises an independently driven wheel with an in-wheel motor, a thigh leg with an electric cylinder for actuation, and a calf leg driven by a joint motor with a harmonic drive reduction. However, controlling WLRs poses significant challenges, including: **a**. High ratio of unsprung mass to sprung mass: This can adversely impact the stability and performance of the vehicle. **b**. Terrain Adaptability: WLRs often need to traverse complex and uneven terrains where traditional wheeled or legged robots may struggle. **c**. Stability and Maneuverability: Maintaining stability and maneuverability is crucial for mobile robots, particularly in dynamic and variable environments.

In this paper, we proposed a solution inspired by animals with tails, such as lizards [12] and galagos [13], which use their tails to adjust their body posture during jumping. To achieve this, we develop a state-based model predictive control (MPC) controller that collaborates the thigh leg and calf leg to maintain a stable tire-contact force and ensure ride comfort. The control effect of our method can be briefly illustrated in Fig. 2. When the vehicle encounters a bump, the redundant actuation, in this case, the calf leg, swings down to generate a reaction torque to the wheel, thereby alleviating the increase in the vertical tire force due to the bump, as shown in the left part of the figure. On the other hand, when the vehicle encounters a gully, the calf leg swings up to generate a reaction torque to the wheel, further enhancing the vertical tire force, as shown in the right part of the figure. Through the cooperative control of the thigh leg and calf leg, we can achieve modulation of the vertical tire force and stabilization of the sprung mass when the WLR glides over rough terrain.

The rest of this paper is organized as follows. Section II introduces the mathematical model of the wheel-legged system. The development of the state-based MPC controller as well as the control architecture are highlighted in Section III. The simulation verification of the proposed state-based MPC controller are presented in Section IV. Section V presents the paper's conclusions

II. Mathematical Model

A. Modeling of Wheel-Legged System

While the wheel-legged robot in wheeled motion, both thigh and calf will collapse to equilibrium position. Unlike in legged motion mode, the swinging angle of legs, especially thigh leg, is relatively small. Thus part of the wheel-leg system can be seen linear, and schematic diagram of the simplified physical quarter-car model of wheel-leg system is shown in Fig.3.

By performing the differentiation required in the Lagrange equation and assuming joint angle variation is small(which means $\cos(\theta)=1, \sin(\theta)=\theta, (\dot{\theta})^2=0$), we get:

$$\left\{\begin{array}{l} m_s\ddot{z}_s = k_s\left(z_t-z_s\right)+c_s\left(\dot{z}_t-\dot{z}_s\right)+Q_s, \\ \quad\left(m_c+m_w\right)\ddot{z}_t+m_c l\ddot{\theta} = -k_s\left(z_t-z_s\right)-c_s\left(\dot{z}_t-\dot{z}_s\right) \\ \qquad\qquad\qquad +k_t\left(z_g-z_t\right)+c_t\left(\dot{z}_g-\dot{z}_t\right)-Q_s, \\ m_c l\ddot{z}_w+m_c l^2\ddot{\theta} = Q_\theta - l m_c \mathrm{g}, \end{array}\right. \tag{1}$$

where, m_s, m_w and m_c are the sprung, wheel and calf leg masses, k_s, k_t, c_s and c_t are respectively the effective stiffness and damping of the suspension and tire. Q_s and Q_θ are the input of suspension and calf joint, as is shown in Fig.3. Define the following state vector: $\boldsymbol{x}=\left[z_s,\dot{z}_s,z_w,\dot{z}_w,\theta,\dot{\theta}\right]^{\mathrm{T}}$, control input: $\boldsymbol{u}=[Q_s, Q_\theta - l m_c g]$, road disturbance: $\boldsymbol{w}=[w_g,\dot{w}_g]$. Noticed that our calf terms keep the $l m_c \mathrm{g}$ in consideration, instead of neglect the gravity terms. Leads to the calf moment is always divided into a part of the balance gravity, which also results in the influence of the counterclockwise calf moment on the system is less than that of the clockwise one.

The states space form of the dynamic model1 can be given as

$$\dot{\boldsymbol{x}} = \widetilde{\boldsymbol{A}}\boldsymbol{x}+\widetilde{\boldsymbol{B}}_u\boldsymbol{u}+\widetilde{\boldsymbol{B}}_w\boldsymbol{w} \tag{2}$$

with

$$\widetilde{\boldsymbol{A}}=\boldsymbol{M}^{-1}\left[\begin{array}{cccccc} 0 & 1 & 0 & 0 & 0 & 0 \\ -k_s & -c_s & k_s & c_s & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ k_s & c_s & -\left(k_s+k_t\right) & -\left(c_s+c_t\right) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right],$$

$$\widetilde{\boldsymbol{B}}_u=\boldsymbol{M}^{-1}\left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{array}\right], \widetilde{\boldsymbol{B}}_w=\boldsymbol{M}^{-1}\left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ k_t & c_t \\ 0 & 0 \\ 0 & 0 \end{array}\right]$$

and,

$$\boldsymbol{M}=\left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & m_s & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & m_t+m_b & 0 & m_b l \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & m_b l & 0 & m_b l^2 \end{array}\right]$$

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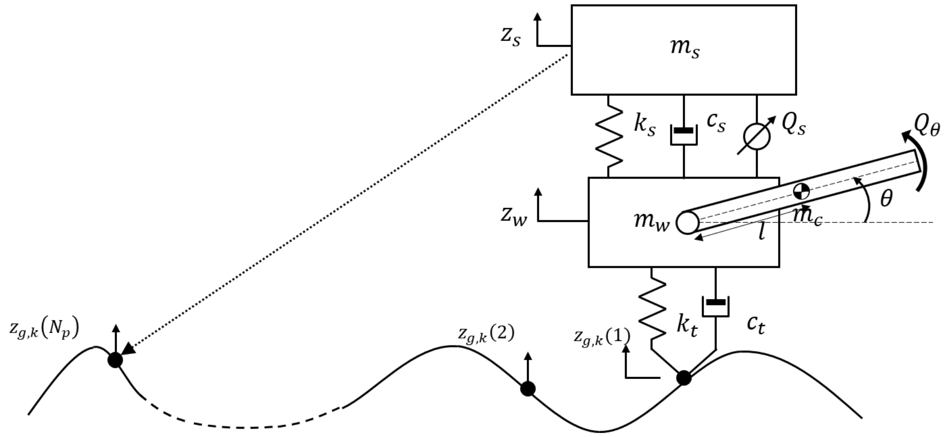

Fig. 3. Simplified wheel-leg system, including thigh and calf actuator, the road profile model and it's preview

When predicting the continuous controlled system state, it is necessary to discretize the controlled object before the controller design:

$$\boldsymbol{x}(k+1) = \boldsymbol{A}\boldsymbol{x}(k) + \boldsymbol{B}_u\boldsymbol{u}(k) + \boldsymbol{B}_w\boldsymbol{w}(k) \tag{3}$$

the equation (3) is the discretized form of (2), where,

$$\boldsymbol{A} = \mathrm{e}^{\widetilde{\boldsymbol{A}}t_s}, \boldsymbol{B}_u = \int_0^{t_s} \mathrm{e}^{\widetilde{\boldsymbol{A}}t}\mathrm{d}t\widetilde{\boldsymbol{B}}_u, \text{and } \boldsymbol{B}_w = \int_0^{t_s} \mathrm{e}^{\widetilde{\boldsymbol{A}}t}\mathrm{d}t\widetilde{\boldsymbol{B}}_w.$$

B. Problem Statement

Generally, the wheel-legged robot can be regarded as a vehicle with rocker-arm suspension [14] or trailing-arm suspension [15] (perpendicular to and forward of the virtual axle) during wheeled motion. The suspension performances concerned include ride comfort and road holding ability whether there is passenger or not. As for the mechanical structure of the robot, mechanical constraints should also be concerned. Thus, the following performance requirements are considered:

(1) *Ride comfort*: ride comfort is a key indicator of the performance for traditional vehicle. For the unmanned robot going forward at high speed in the off-road situation, ground fluctuation may adversely affect the on-board sensors and precision instruments. Thus, the vertical acceleration of the sprung mass $\ddot{z}_s$ should be minimized to reduce vibration of robot body.

(2) *Road holding ability*: Road holding ability can directly affect the power performance, braking performance, and handling stability of the robot since the tire is the only interaction between the wheel-legged robot and the ground during wheeled motion. Therefore, the vertical tire-road contact force should be stable and not affected by a rugged road surface.

(3) *Mechanical constraints*: The limited working space of the thigh and calf leg of the wheel-legged robot can cause the calf to collide with the ground and thigh mechanism during wheeled motion, resulting in damage to the mechanism and actuator and irreparable consequences. Additionally, saturation of the joint actuators may cause control performance deterioration and even instability of the controlled system. Thus, mechanical constraints need to be considered in the controller design.

In the following section, we will design the preview controller to improve both ride comfort performance and road holding ability while considering the mechanical constraints.

C. Shift Register for Road Preview

For preview control, the vertical road profile is modeled through a shift register and derivative of the profile is calculated by the buffered profile. Let the t_s be the sampling time and N_p be the prediction length of the register. Then the previewer is represented in discrete time form as

$$\begin{bmatrix} w_{k+1}(N_p) \\ w_{k+1}(N_p-1) \\ \vdots \\ w_{k+1}(2) \\ w_{k+1}(1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} w_k(N_p) \\ w_k(N_p-1) \\ \vdots \\ w_k(2) \\ w_k(1) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} z_{g,k} \tag{4}$$

$$\begin{bmatrix} \dot{w}_{k+1}(N_p) \\ \dot{w}_{k+1}(N_p-1) \\ \vdots \\ \dot{w}_{k+1}(2) \\ \dot{w}_{k+1}(1) \end{bmatrix} = \frac{1}{t_s} \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & -1 \end{bmatrix} \begin{bmatrix} w_k(N_p) \\ w_k(N_p-1) \\ \vdots \\ w_k(2) \\ w_k(1) \end{bmatrix} + \frac{1}{t_s} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} z_{g,k}, \tag{5}$$

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where, $z_{g,k}$ is the current road profile sensed by preview sensor. And we combined the road input sequence $\boldsymbol{w}(k) = [\boldsymbol{w}_k, \dot{\boldsymbol{w}}_k]^{\mathrm{T}}$, which has a length of $2 \times N_p$, for the system state prediction.

III. Controller Design

A. Return control

The controller design begins with a return controller, which is the simplest control strategy for WLRs. In this approach, the suspension travel is maintained at a predetermined length, and the joint motor locks the calf at a fixed angle relative to the thigh and the ground. The return controller is employed to restore the thigh and calf legs to their equilibrium positions. A Proportional-Integral (PI) controller can be utilized to implement the return controller:

$$\begin{aligned} Q_s^{\mathrm{Re}} &= k_p(z_{\mathrm{eq}} - z) + k_i \int (z_{\mathrm{eq}} - z)\mathrm{d}t, \\ Q_\theta^{\mathrm{Re}} &= k_p(\theta_{\mathrm{eq}} - \theta) + k_i \int (\theta_{\mathrm{eq}} - \theta)\mathrm{d}t, \end{aligned} \tag{6}$$

where k_p and k_i represent the proportional and integral gains of the PI controller, respectively. z is the difference between the suspension travel z_s and the wheel height z_w, and z_{eq} is a reference value for the suspension travel. θ_{eq} is a predetermined calf joint angle. A relatively large integral gain is chosen in order to introduce a delayed response characteristic that allows the system to gradually return to equilibrium without immediately affecting the preview controller.

B. System Output

As mentioned previously, it is imperative for the robot to maintain an acceleration of the sprung mass that approaches zero and a relatively stable dynamic tire load. Let $\boldsymbol{y}^{\mathrm{ref}}(k) = \left[\dddot{z}^{\mathrm{ref}}s(k), F_{tire}^{\mathrm{ref}}(k)\right]^{\mathrm{T}}$ denote the reference values for sprung mass acceleration and vertical tire force, respectively. In each time step, the reference sprung mass acceleration is set to $\dddot{z}^{\mathrm{ref}}s(k) = 0$, and the reference vertical tire force is given by $F_{tire}^{\mathrm{ref}}(k) = (m_s + m_w + m_c)\mathrm{g}$. Here, we use the vertical tire force as a representation of the road holding ability of the robot, instead of directly using the dynamic tire load, for several reasons. Firstly, the wheel needs to roll for a certain distance and there is a time delay before the lateral and longitudinal forces of the tire are fully generated. However, the reaction force of the machine leg is not affected by this delay, hence the use of vertical tire force as a proxy.

Another important consideration is that the robot operates on highly uneven road conditions, which can result the system in two distinct states: the stance state and the flight state. In the stance state, the robot has ground support and at least one foot is in contact with the ground, while in the flight state, the wheel is off the ground. This differentiation is crucial as it affects the control strategy and dynamics of the robot, requiring different objects to control.

The output function can be given by:

$$\boldsymbol{y}(k) = \boldsymbol{C}\boldsymbol{x}(k) + \boldsymbol{D}_u\boldsymbol{u}(k) + \boldsymbol{D}_w\boldsymbol{w}(k) \tag{7}$$

In order to minimize the changes in the reference trajectories of the MPC controller, two sets of output matrices and direct transmission matrices are defined based on the system states. The mathematical expressions for these matrices are given below:

$$\begin{aligned} &\boldsymbol{C}^{\mathrm{stance}} = \begin{bmatrix} -\frac{k_s}{m_s} & -\frac{c_s}{m_s} & \frac{k_s}{m_s} & \frac{c_s}{m_s} & 0 & 0 \\ 0 & 0 & -k_t & -c_t & 0 & 0 \end{bmatrix}, \\ &\boldsymbol{D}_u^{\mathrm{stance}} = \begin{bmatrix} \frac{1}{m_s} & 0 \\ 0 & 0 \end{bmatrix}, \boldsymbol{D}_w^{\mathrm{stance}} = \begin{bmatrix} 0 & 0 \\ k_t & c_t \end{bmatrix}, \end{aligned} \tag{8}$$

and, if η is a scale factor which satisfies $\eta > 0$, the output matrix and direct transmission matrices in flight state are

$$\begin{aligned} &\boldsymbol{C}^{\mathrm{flight}} = \begin{bmatrix} -\frac{k_s}{m_s} & -\frac{c_s}{m_s} & \frac{k_s}{m_s} & \frac{c_s}{m_s} & 0 & 0 \\ -\frac{k_s}{\eta} & -\frac{c_s}{\eta} & \frac{k_s}{\eta} & \frac{c_s}{\eta} & 0 & 0 \end{bmatrix}, \\ &\boldsymbol{D}_u^{\mathrm{flight}} = \begin{bmatrix} \frac{1}{m_s} & 0 \\ \frac{1}{\eta} & \frac{1}{\eta l} \end{bmatrix}, \boldsymbol{D}_w^{\mathrm{flight}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned} \tag{9}$$

The choice of output matrix and direct transmission matrix in equation (7) is dependent on the finite state machine (FSM):

$$\begin{aligned} &\begin{cases} \boldsymbol{C} = \boldsymbol{C}^{\mathrm{stance}}, & \text{if } z_w < z_g \text{ or } \dot{z}_w < \dot{z}_g \\ \boldsymbol{C} = \boldsymbol{C}^{\mathrm{flight}}, & \text{if} z_w \geq z_g \text{ and } \dot{z}_w \geq \dot{z}_g \end{cases} \\ &\begin{cases} \boldsymbol{D}_u = \boldsymbol{D}_u^{\mathrm{stance}}, & \text{if } z_w < z_g \text{ or } \dot{z}_w < \dot{z}_g \\ \boldsymbol{D}_u = \boldsymbol{D}_u^{\mathrm{flight}}, & \text{if} z_w \geq z_g \text{ and } \dot{z}_w \geq \dot{z}_g \end{cases} \\ &\begin{cases} \boldsymbol{D}_w = \boldsymbol{D}_w^{\mathrm{stance}}, & \text{if } z_w < z_g \text{ or } \dot{z}_w < \dot{z}_g \\ \boldsymbol{D}_w = \boldsymbol{D}_w^{\mathrm{flight}}, & \text{if} z_w \geq z_g \text{ and } \dot{z}_w \geq \dot{z}_g \end{cases} \end{aligned} \tag{10}$$

In both (8) and (9), the first row of (7) corresponds to the observation of the sprung mass vertical acceleration. The second row of (7) pertains to the observation of the vertical tire force in the stance state. Additionally, in the second row of (9), the purpose of constructing the observer is to facilitate the wheel's return to the ground with an acceleration of ηg, while in stance state as (8), the steady-state vertical tire force F_{tire}^{ref} tracking carries the physical meaning of the wheel-leg system returning to the ground with an acceleration of g.

C. System Prediction

System prediction serves as the foundation for the controller with preview, aiming to anticipate the future state and output of the system based on the current system state and the input of the future road preview. When the control horizon is denoted as N_c and prediction horizon is denoted as N_p, the future state and output of the wheel-leg system can be calculated as follows:

$$\begin{aligned} \hat{\boldsymbol{x}} &= \boldsymbol{\Psi}\hat{\boldsymbol{\xi}} + \boldsymbol{\Omega}\hat{\boldsymbol{u}} \\ \hat{\boldsymbol{y}} &= \boldsymbol{\Lambda}\hat{\boldsymbol{\xi}} + \boldsymbol{\Theta}\hat{\boldsymbol{u}} \end{aligned} \tag{11}$$

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with,

$$\boldsymbol{\Psi}=\begin{bmatrix} \boldsymbol{A} & \boldsymbol{B}_w & 0 & \cdots & 0 \\ \boldsymbol{A}^2 & \boldsymbol{A}\boldsymbol{B}_w & \boldsymbol{B}_w & \cdots & 0 \\ \boldsymbol{A}^3 & \boldsymbol{A}^2\boldsymbol{B}_w & \boldsymbol{A}\boldsymbol{B}_w & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}^{N_p} & \boldsymbol{A}^{N_p-1}\boldsymbol{B}_w & \boldsymbol{A}^{N_p-2}\boldsymbol{B}_w & \cdots & \boldsymbol{B}_w \end{bmatrix},$$

$$\boldsymbol{\Omega}=\begin{bmatrix} \boldsymbol{B}_u & 0 & 0 & \cdots & 0 \\ \boldsymbol{A}\boldsymbol{B}_u & \boldsymbol{B}_u & 0 & \cdots & 0 \\ \boldsymbol{A}^2\boldsymbol{B}_u & \boldsymbol{A}\boldsymbol{B}_u & \boldsymbol{B}_u & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{A}^{N_p-1}B\boldsymbol{B}_u & \boldsymbol{A}^{N_p-2}\boldsymbol{B}_u & \boldsymbol{A}^{N_p-3}\boldsymbol{B}_u & \cdots & \boldsymbol{B}_u \end{bmatrix},$$

$$\boldsymbol{\Lambda}=\begin{bmatrix} \boldsymbol{C} & \boldsymbol{D}_w & 0 & \cdots & 0 \\ \boldsymbol{C}\boldsymbol{A} & \boldsymbol{C}\boldsymbol{B}_w & \boldsymbol{D}_w & \cdots & 0 \\ \boldsymbol{C}\boldsymbol{A}^2 & \boldsymbol{C}\boldsymbol{A}\boldsymbol{B}_w & \boldsymbol{C}\boldsymbol{B}_w & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{C}\boldsymbol{A}^{N_p-1} & \boldsymbol{C}\boldsymbol{A}^{N_p-)}\boldsymbol{B}_w & \boldsymbol{C}\boldsymbol{A}^{N_p-3}\boldsymbol{B}_w & \cdots & \boldsymbol{D}_w \end{bmatrix}$$

$$\boldsymbol{\Theta}=\begin{bmatrix} \boldsymbol{D}_u & 0 & \cdots & 0 \\ \boldsymbol{C}\boldsymbol{B}_u & \boldsymbol{D}_u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{C}\boldsymbol{A}^{N_c-2}B_u & \boldsymbol{C}\boldsymbol{A}^{N_c-3}\boldsymbol{B}_u & \cdots & \boldsymbol{D}_u \\ \boldsymbol{C}\boldsymbol{A}^{N_c-1}B_u & \boldsymbol{C}\boldsymbol{A}^{N_c-2}\boldsymbol{B}_u & \cdots & \boldsymbol{C}B_u \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{C}\boldsymbol{A}^{N_p-2}B_u & \boldsymbol{C}\boldsymbol{A}^{N_p-3}\boldsymbol{B}_u & \cdots & \boldsymbol{C}\boldsymbol{A}^{N_p-N_c-1}\boldsymbol{B}_u \end{bmatrix}$$

and,

$$\hat{\boldsymbol{x}}=\begin{bmatrix} \boldsymbol{x}(k+1) \\ \boldsymbol{x}(k+2) \\ \vdots \\ \boldsymbol{x}(k+N_p) \end{bmatrix}, \hat{\boldsymbol{y}}=\begin{bmatrix} \boldsymbol{y}(k) \\ \boldsymbol{y}(k+1) \\ \vdots \\ \boldsymbol{y}(k+N_p-1) \end{bmatrix},$$

$$\hat{\boldsymbol{u}}=\begin{bmatrix} \boldsymbol{u}(k+1) \\ \boldsymbol{u}(k+2) \\ \vdots \\ \boldsymbol{u}(k+N_c) \end{bmatrix}, \hat{\boldsymbol{w}}=\begin{bmatrix} w(k+1) \\ w(k+2) \\ \vdots \\ w(k+N_p) \end{bmatrix}, \hat{\boldsymbol{\xi}}=\begin{bmatrix} \boldsymbol{x} \\ \hat{\boldsymbol{w}} \end{bmatrix}.$$

D. QP Problem Formulation

The desired system outputs in the prediction horizon N_p and external commands in the control horizon N_c are incorporated into a cost function as follows:

$$J(\hat{\boldsymbol{y}},\hat{\boldsymbol{u}},\varepsilon)=\sum_{i=k}^{k+N_p-1}\left\|\boldsymbol{y}-\boldsymbol{y}^{\text{ref}}\right\|_{\boldsymbol{Q}}^2+\sum_{i=k+1}^{k+N_c}\|u(k)\|_{\boldsymbol{R}}^2+\rho\varepsilon^2, \tag{12}$$

where, $\boldsymbol{Q}\in\mathbb{R}_{2N_p\times 2N_p}$ and $\boldsymbol{R}\in\mathbb{R}_{2N_c\times 2N_c}$ are the positive-definite weight matrices:

$$\boldsymbol{Q}=\begin{bmatrix} \boldsymbol{q}_1 & 0 & \cdots & 0 \\ 0 & \boldsymbol{q}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{q}_{N_p} \end{bmatrix}, \boldsymbol{R}=\begin{bmatrix} \boldsymbol{r}_1 & 0 & \cdots & 0 \\ 0 & \boldsymbol{r}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{r}_{N_c} \end{bmatrix}.$$

In predictive control, the control input at the first time step is generally more concerned, so the attenuation factor can be used between each weight factor $\boldsymbol{q},\boldsymbol{r}\in\mathbb{R}_{2\times 2}$:

$$\begin{aligned} \alpha_q^{N_p-1}\boldsymbol{q}_1=\alpha_q^{N_p-2}\boldsymbol{q}_2=\cdots=\boldsymbol{q}_{N_p}, \\ \alpha_r^{N_c-1}\boldsymbol{r}_1=\alpha_r^{N_c-2}\boldsymbol{r}_2=\cdots=\boldsymbol{r}_{N_c}. \end{aligned} \tag{13}$$

Notice in (12), a relaxation factor ε and its associated weight factor ρ are introduced as a new component of the control input. By adding $\varepsilon\geq 0$ to the upper bound of constraints and subtracting it from the lower bound, the hard joint constraints can be converted into soft constraints, allowing for the possibility of infeasible solutions. A relatively large weight ρ and state prediction horizon N_p can enable the controller to react promptly in case of potential mechanical collisions.

Write (12) in a compact form:

$$J(\hat{\boldsymbol{y}},\hat{\boldsymbol{u}},\varepsilon)=\left[\hat{\boldsymbol{u}}^{\mathrm{T}},\varepsilon\right]^{\mathrm{T}}\boldsymbol{H}\left[\hat{\boldsymbol{u}}^{\mathrm{T}},\varepsilon\right]+\boldsymbol{G}\left[\hat{\boldsymbol{u}}^{\mathrm{T}},\varepsilon\right]^{\mathrm{T}}, \tag{14}$$

where,

$$\boldsymbol{H}=\begin{bmatrix} \boldsymbol{\Theta}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{\Theta}+\boldsymbol{R} & 0 \\ 0 & \rho \end{bmatrix}, \boldsymbol{G}=\begin{bmatrix} 2(\boldsymbol{y}-\boldsymbol{y}_{ref})^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{\Theta} \end{bmatrix}.$$

For each step, the predictive controller finds the optimal sequence of the control input $\hat{\boldsymbol{u}}$ over N_c that minimizes the cost function. The QP problem can then be formulated as follows:

$$\begin{array}{lll} \min\limits_{\hat{\boldsymbol{u}},\varepsilon} & \left[\hat{\boldsymbol{u}}^{\mathrm{T}},\varepsilon\right]^{\mathrm{T}}\boldsymbol{H}\left[\hat{\boldsymbol{u}}^{\mathrm{T}},\varepsilon\right]+\boldsymbol{G}\left[\hat{\boldsymbol{u}}^{\mathrm{T}},\varepsilon\right]^{\mathrm{T}} & \\ s.t. & \boldsymbol{u}(k+i)\leq\boldsymbol{u}_{\max}, & i=1,2,3\ldots,N_c \\ & \boldsymbol{x}(k+j)\leq\boldsymbol{x}_{\max}+\varepsilon, & j=1,2,3\ldots,N_p \\ & -\boldsymbol{u}(k+i)\leq-\boldsymbol{u}_{\min}, & i=1,2,3\ldots,N_c \\ & -\boldsymbol{x}(k+j)\leq-\boldsymbol{x}_{\min}+\varepsilon, & j=1,2,3\ldots,N_p \\ & -\varepsilon\leq 0 & \end{array}. \tag{15}$$

The problem stated in (15) is a constrained quadratic programming (QP) problem that can be solved using a QP solver. Once solved, the first term of $\hat{\boldsymbol{u}}$ over the prediction horizon N_c will be utilized as the actuator input in the real-time implementation.

E. Overall Controller Architecture

Both the MPC and return controller components are integrated into the preview controller, which is named the preview controller in this study. The MPC being responsible for immediate control responses to road disturbances and the return controller for slowly returning the system to equilibrium. A preview sensor is utilized to collect the current road profile information z_g, which is stored in the shift register $\boldsymbol{\omega}$. The FSM sends the system state information based on the vertical displacement of the wheel and ground. The MPC module utilizes the constant predetermined control reference $\boldsymbol{y}^{\text{ref}}$ and the estimated or directly sensed system states to formulate a QP problem, which is then solved by a QP solver to obtain an optimal solution sequence $\hat{\boldsymbol{u}}$. The first item of the sequence, combined with the return control results $\boldsymbol{Q}^{\text{Re}}$, is then sent to the actuation system of the WLR, which consists of an electric cylinder for the thigh leg and a joint motor for the calf leg.

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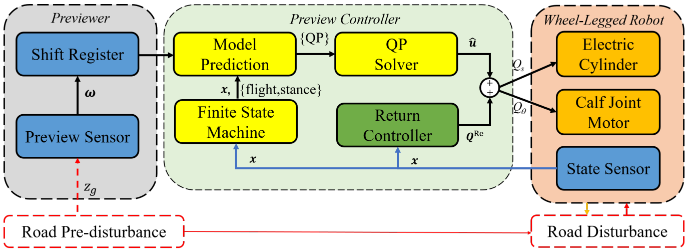

Fig. 4. The overview of the preview controller architecture

TABLE I
SIMULATION PARAMETERS

Parameters	Units	Values	Parameters	Units	Values
Sprung mass	kg	40	Wheel mass	kg	30
Thigh mass	kg	20	Wheel radius	cm	26.67
Thigh length	m	0.6	Tire stiffness	N/m	11768
Calf mass	kg	30	Tire damping	N/(m/s)	1176.8
Calf length	m	0.4	Thigh equilibrium	deg	-45
Gear ratio		10	Calf equilibrium	deg	135
Motor inertia	kg·m^2	0.029	Calf constraint	deg	$\pm$ 20

IV. Simulation Results

To verify the feasibility of the collaborative wheel leg control algorithm proposed in this study, a simulation platform was developed using the *Simscape Multibody*, as depicted in Fig. 5. The simulation platform includes a quarter body model, thigh leg, calf leg, wheel, tire, and ground model. In order to simulate the behavior of the electric cylinder, a shock absorber with a prismatic joint was utilized, with the stiffness and damping of the prismatic joint set at 16000 N/m and 1000 N/(m/s), respectively, to emulate the self-locking characteristic of an electric cylinder. The drive from the thigh leg directly acts on the absorber, while the lower leg is driven by a gear reducer that simulates the action of a driving motor through internal meshing gears. As the simulation is conducted solely to verify the principle, only basic rigid body elements with uniform mass distribution are used.

The interaction between the tire and the ground was modeled using the *Simscape Multibody Contact Forces Library*, while neglecting ground deformation. The wheel leg system is connected to the world coordinate system via a planar pair, assuming that the system does not experience pitch, roll, yaw, or lateral movement of the vehicle, and only considers longitudinal and vertical movements. The QP problem in the preview controller is solved using the *quadprog* function in *MATLAB*. The simulation step size is set at 0.005s, and other simulation parameters are detailed in Table I.

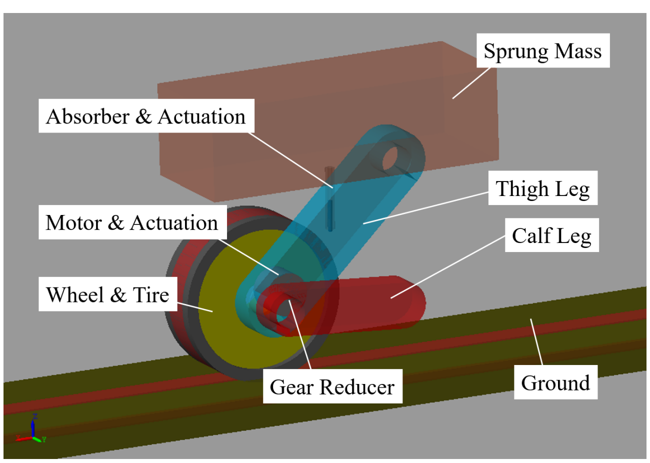

Fig. 5. Simulation model of quarter wheel-legged system in Simscape Multibody environment

To evaluate the effectiveness of the preview controller, we compare it with a single return controller, as described in section III. Prior to the simulation, the wheel-leg system is suspended in the air with a small distance between the wheel and the ground, and undergoes free-falling motion at the beginning of the simulation. Subsequently, the steady-state state of the quarter-car model is recorded from 0.12 to 4 seconds. At 4 seconds, sinusoidal excitation is applied to the ground as shown in Fig. 6. The excitation continues for 5 seconds before ending, and the system returns to a steady-state standing posture. The obtained results are presented below:

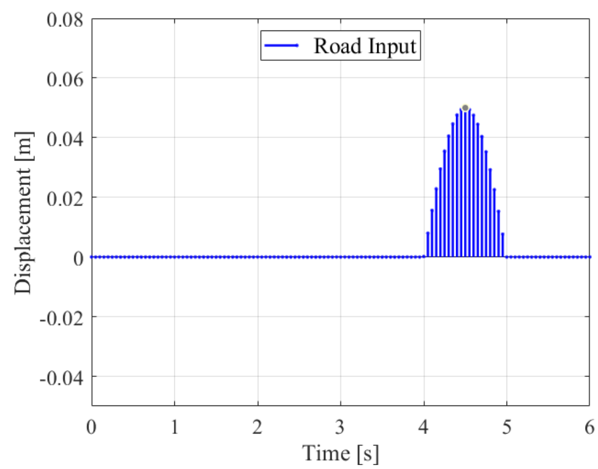

Fig. 6. Road input in the simulation environment

Upon comparing the control objectives of body vertical acceleration and tire vertical force in Fig. 7 and Fig. 8, it is evident that the use of the preview controller effectively reduces the peak vertical acceleration of the body and stabilizes the vertical force of the tire, while adhering to the mechanical constraints of the calf leg. The oscillation observed in vertical accelerations and tire forces before 4 seconds is attributed to the relatively high gains of the return controller, which is utilized in both the proposed preview controller and the comparison term to maintain system stability. The tire forces and body acceleration exhibit two surges at the onset and completion of the ground excitation at 4 seconds and 5 seconds, respectively. These abrupt changes may be attributed to the intermittent nature of the sinusoidal road excitation and the contact mechanism in the simulation environment, which may not respond rapidly enough to changes in road

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input. However, even with these surge phenomena, the MPC controller with preview information significantly mitigates these effects.

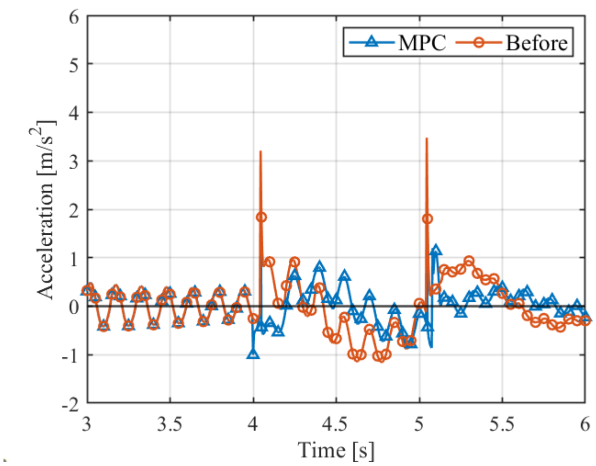

Fig. 7. Comparison of vertical accelerations of the sprung mass

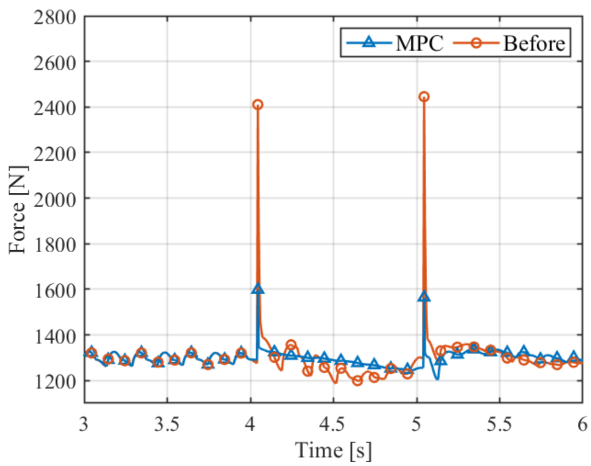

Fig. 8. Comparison of vertical tire forces

The heights of the wheel centers for both controllers are presented in Fig. 9. Despite the original intention of incorporating falling-down control for the wheel-legged system during the flight phase, achieving precise falling speed control at arbitrary heights remains challenging due to mechanical constraints, particularly the limited workspace of the calf joint angle. As a result, we only validate the ability to adjust falling speed at a restricted initial height. In this study, we utilize the height of the wheel centers to assess the contact state between the wheel and the ground. Notably, the preview controller exhibits earlier recovery from falling compared to the comparison term, as evident from the enlarged height figure. Frankly, there is no significant reduction in contact time, as the falling phase of the wheel leg system is already brief, lasting less than 0.1 seconds.

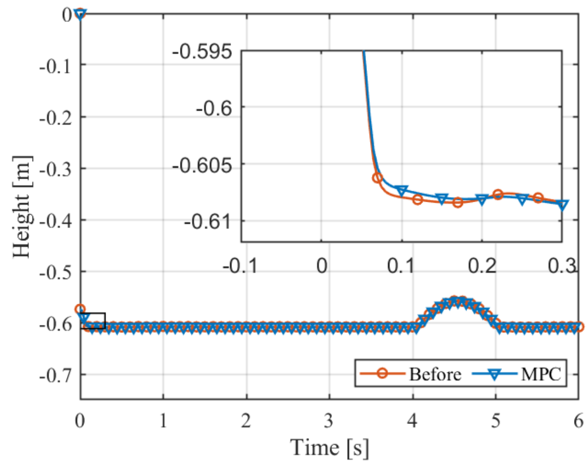

Fig. 9. Heights of the wheels' center in simulation environment

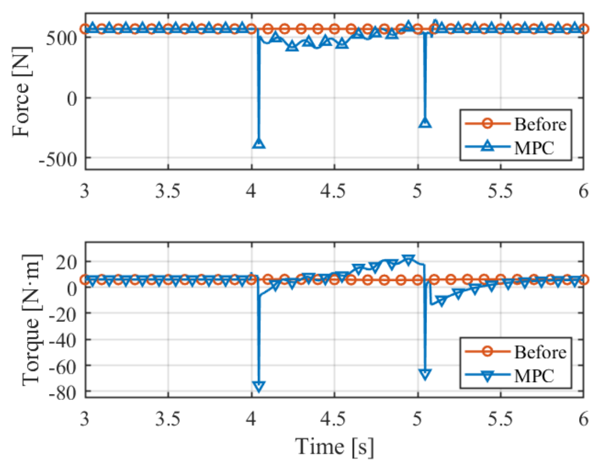

Fig. 10. Comparison of thigh joint actuation (upper) and calf joint torques (lower)

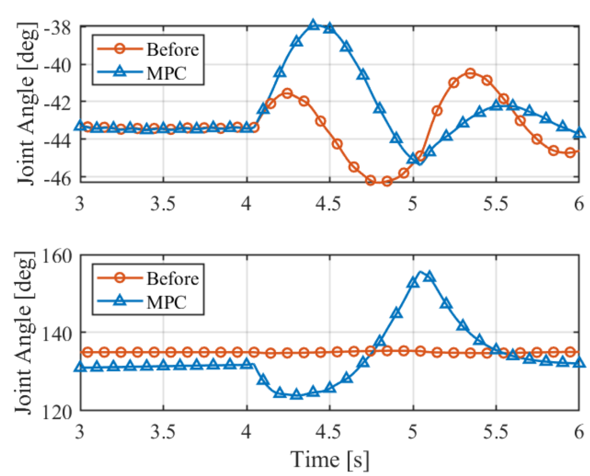

Fig. 11. Changes of the thigh joint (upper) and calf joint (lower) during the simulation

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Fig. 10 presents a comparison of the actuations between the two controllers, where the torques of the motor before the reducer are used, instead of the effective calf joint torque. It is observed that both the thigh and calf actuations exhibit sudden changes, which are aligned with the abrupt variations in tire force and vertical acceleration of the body, in order to achieve the control objectives. However, it should be noted that these actuations can only be realized in a simulation environment, as input variations during controller design are not limited, making it challenging to achieve such rapid changes in actuation for both the joint motor and electric cylinder in real-world settings. Additionally, obtaining timely and accurate preview of road input is also challenging in practice. Nevertheless, these findings are sufficient to verify that the collaborative control of the thigh and calf legs can indeed improve ride comfort and road-holding ability.

Fig. 11 illustrates the process of changing the angles of the thigh and calf legs. In our coordinate system definition, the vertical downward direction of the thigh leg and the vehicle's rotating joint is defined as 0°, with the steady-state angle set at -45°. The extended line facing the thigh leg represents the 0° line of the calf leg rocker arm, with the steady-state angle set at 135°. As we have only imposed constraints on the calf joint, which is more likely to come into contact with the ground, the changes in calf joint angle demonstrate that our controller is able to operate within the specified constraints. When comparing the proposed preview controller with the return controller, it can be observed that the leg angles exhibit more variability when encountering road disturbances, and the movement of both the thigh and calf legs align with the working principle explained in Fig. 2.

V. Conclusion

We proposed a collaborative controller method based on MPC with road preview for knee-wheeled legged robots to address the control contradiction of vertical comfort and road holding ability, while considering mechanical constraints. Our approach incorporates the calf as a redundant component into the control algorithm, utilizing its movement to improve the performance of the wheel-leg system. The wheel-leg mechanism of the quarter model is linearized at the working equilibrium point, and a preview controller is constructed. Simulation results using Simscape Multibody validated the effectiveness of the proposed control algorithm in reducing peak vertical acceleration of the robot body and obtaining smoother tire vertical force compared with a PI controller which only lead the system return to the equilibrium state. The simulation results demonstrated that a collaborative control approach utilizing road preview can effectively improve the robot's overall performance in terms of both comfort and road holding ability. The insights gained from this study can inform the development of control strategies for knee-wheeled legged robots and advance research in wheel-legged locomotion.

In our future work, we have several planned tasks. First, we will extend the quarter car model to the entire vehicle model in order to evaluate additional performance indicators for pitch and roll at the center of mass of the robot body. This will provide a more comprehensive understanding of the overall performance of the robot in different dynamic scenarios. Second, we will explore treating the road input as an unknown disturbance, rather than relying solely on the road preview sensor, in order to achieve similar performance in variable road conditions. This approach may provide more robustness and adaptability to changing environments. Third, we will investigate the combination of handling stability control with the proposed suspension controller, taking into consideration the possibility of four wheels being off the road. This will allow us to address handling stability issues in extreme driving conditions.

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